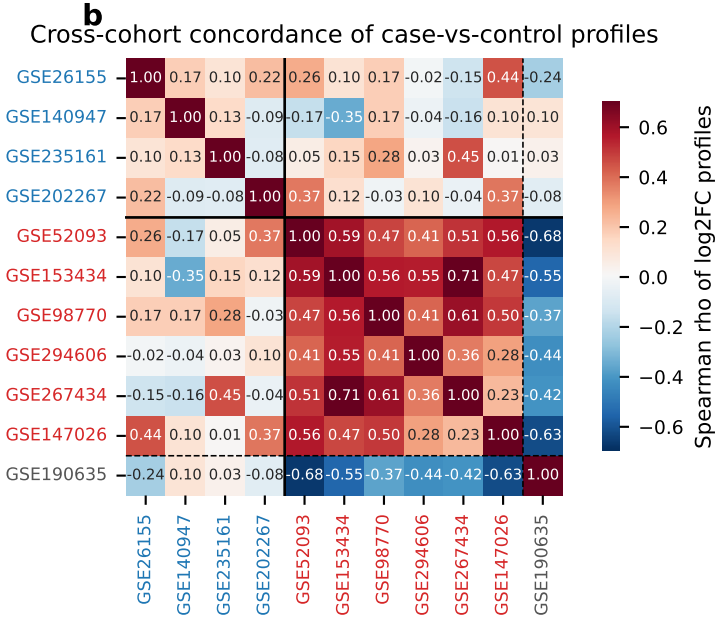


a

Data structure	Unit of analysis	Inference status (plan lock after exploration)
Bulk transcriptomes 10 cohorts: 4 ATAA (63 vs 52), 6 TAAD (41 vs 35) + GSE190635 (flagged, sensitivity only)	Unit: patient within cohort; cohorts compared at cohort level	F1 primary: C, I, R (Holm) TAAD effect vs ATAA effect
scRNA-seq, patient x cell-type pseudo-bulk GSE155468 (ATAA 8 vs 3; 7 vs 3 with >= 20 contractile SMC), GSE213740 (TAAD 6 vs 3), GSE189795 (acute TAAD 5 vs 4; no contractile-SMC cluster recovered)	Unit: patient (>= 20 cells of the type) label-free scores within cell type	F2 primary: R in contractile SMC (exact enumeration; one test per dataset; not estimable if the cell type is missing)
Within-patient regions GSE140947: aneurysm belly vs neck (6 pairs)	Unit: within-patient difference	F3 primary: delta R (exact sign-flip, 64 assignments)
Stage and wall layer GSE222318 (acute/subacute/chronic; sampling position differs) GSE318877 (3 dissection vs 4 dilatation; 4 medial layers)	Unit: patient; patient x layer	Estimation only (no inferential test; CI and raw scores)
Ordered dilatation within one cohort GSE26155: No (31) < Borderline (6) < Yes (22)	Unit: patient (intima-media, TAV)	Exploratory trend (Jonckheere-Terpstra, permutation P; BH)

Programme scores (Tier 1, external definitions): C = SMC contractile module; I = mean of six injury-response modules (hypoxia, glycolysis, oxidative stress/NFE2L2, MYC-ribosome, p53-DNA damage, NF-kB/IL-6); R = I + C



c

Reproducibility of gene-level findings (REM primary)

	ATAA (4 cohorts)	TAAD (6 cohorts)	TAAD 4-of-6 subsets
REM consensus DEGs	171	1917	1305-1992
LOCO retention of consensus (median)	0.28	0.74	0.42 (min)
Median pairwise rho	0.12	0.51	0.38-0.57
Consensus under HKSJ	0	354	-
Disease-difference genes (enriched in ATAA / TAAD)	34	778	27 shared 9 discordant

Gene-level asymmetry is a statement about statistical reproducibility across the included cohorts, not about biological stability.